Q-Learning □□ SDN □□□ □□□ □□

A Q-Learning based Routing Optimization Model in a Software Defined Network

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IEEE 19th Int'l Conference on Embedded and Ubiquitous Computing (EUC), 2021

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□□□□	2021

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[□□ □□]

SDN(Software Defined Network) □□□□ □□□ QoS □ □□□□
□□□ □□□ □□□□ □□ , Q-Learning □□□ □□□ □□□□□ □□□□ .

□□□ □ □□□ □□□□ □□□ QoS □□□ □□□ □□□□
□□□ □□ □□ (Reward Function) □ □□□□ □□ □□□ □□□□□ .

[□□ □□]

1. SDN Controller □ Q-Learning □ Agent □ □□
2. □□□□ □□ (State) □ □□ □□ □ (Action) □□
3. □□□ + □□□ □□□ □□□ □□ □□ □□
4. epsilon-greedy □□□□ □□ / □□ □□
5. Q-Table □□ -> □□ □□□ □□□ □□

[□□ □□]

- □□ □□□□ □□□□ □□ :
  □□ □□ □□ **43.38%** □□
  □□ □□ □□ **69.57%** □□

- 30% □□ □□□ □□□□ □□□ □□ □□
- 가 □□□ □□ □ □□□□□

□□ □□ □□

[□□ □□□□□ □□]

1. □□ □□ + □□□ □□□ □□□□□ □□
   -> □□□□ □□ □ □□□□ □□□
2. □□ □□□ QoS □□ □□ □□ □□
3. □□ □□□□ (Shortest Path) □□□□□
   □□□□ □□□ □□□ □□□□ □□
4. 가 □□ □□□□ □□□□□
   □□ □□ □□ □□

[□ □□□ □□□□□ □□ □□]

SDN □□□□
- □□□ □ □□□ (non-identical) □□□□
- □□□ (heterogeneous) QoS □□□ □□□□

□□ □□ (Delay) □ □□□□□□
- □□□ QoS □ □□□□

□□□ □□□ □□□ □□□□ □□
-> Q-Learning □□ □□

□□ □□ : □□□ □□ □□ (Reward Function) □□

□□□ □□ □□ : SDN + Q-Learning

STEP 1

SDN Controller (Agent)

□□□□ □□ □□ □□
(□□□□ , □□□ , □□□)

STEP 2

Q-Learning □□□□

□□□ □□ □□ □□□
□□ □□ (ε-greedy)

STEP 3

Q-Table □□

□□ □□□ □□□
□□ □ □□

[□□ □□□□]

SDN Controller □ Agent □□ □□□□ □□□ □□□□□□ Q-Learning □□ □□
□□□ (Bandwidth) □ □□ □□ (Delay) □ □□□ □□□ □□ □□□ □□ (Q-Table) □□

Q-Table □□ □ , □□□ □□□ Q-Table □ □□ □□□ -> □□ □□ □□□ + QoS □□

□□□□ □□ (1) : Agent & State

Agent (□□□□) = SDN Controller

[□□]
- □□□□ □□ □□ (Global View) □□ □□
- Q-Learning □□ □□□ □□□□ □□
- Flow Table □ □□□□ □□

[□□]
- 가 □□□□□□□□ □□□□
- □□□ □□ □□□□□□ Q-Table □□
- □□ □□ □ □□ □□□□ □□□ □□

State (□□)

[□□] s(h,k) = i, i = [1, 10]

[□□]
- □□ □□□ k 가□□□ □□□ □□ i
- □ (hop) h □□□ □□□□ □□

[□□ □□]
- □□□□ □□□□ □□ Y[i,j]
  (Y[i,j]=1: □□ / Y[i,j]!=1: □□□)
- 가 □□□□ Ba (0~1 □□□)
- □□□ □□ □□□ T

□□□□ □□ (2) : Action & Reward

Action (□□)

[□□] a(h,k) = i, Y[i,h]=1 □□□□□

- □□ □ (hop) □□ □□□ □□□ □□ □□
- 가 □□□ □□□□ □□ □ □□□□□
- □□ : epsilon-greedy (□□ + □□)

Reward (□□) - □□□ □□ □□

[□□ □□ □□] (□□□ □ □□ □□)

R = B1*(1 - delay) + B2*(Bmin/Ba)

[□ , Ba < Bmin □□ R = B3*(Ba - Bmin)]

delay □□□□ , Bmin/Ba □□□□ -> 가 □□

B1, B2 : QoS 가 □ □□(□□ □□)

[□□ □□ □□ □□]

□□ □□ □□□ □□ :

- □□ □ □ □□ □□ □□□ □□
- -> □□ □□□ / □□ □□

□□ □□ □□□ 3 가□□ □□ :

(1) delay □□□ :

delay = T / (SE * Ba)

(T: □□□□ , SE: □□□□□□ , Ba: □□□□□)

(2) □□□ □□□ □□□ :

Bmin/Ba (가 □□ □ □□□□)

(3) 가 □□□□ □□ -> □□ □□

Policy Evaluation - Q □ □□□□

$$Q(s, a) \leftarrow Q(s, a) + a\,[\,R + g * \max Q(s', a') - Q(s, a)\,]$$

a (□□□ , 0<a<1)	□□□ □□□ □□ Q □□ □□□□ □□
g (□□□□ , 0<g<1)	□□ □□□ □□□ g 가 □□□ □□ □□ □□
R(h,k) (□□ □□)	□□ □□□ □□ □□ □ h, □□□ k □ □□
max Q(s', a')	□□ □□□□□ □□ Q □ -> □□ □□ □□□

-> □ □□ □ Q 가ㅅ 가 □□□ □□□ □□□□□ □□ □□

Policy Control - epsilon-greedy □□ □□

[□□ □□ □□]

pi(a | s) = P(a = a* | s(h,k) = s)

epsilon-greedy □□ :
- □□ epsilon -> □□□ □□ (□□)
- □□ 1-epsilon -> □□ □□ (□□)

□□ □□ □□ □ : epsilon = 0.3
(□□□ □□□□□ □□ □□□)

□□ □□ :
Q-Table □ □ □□ □□□ □□ □□□
□□ □□ -> □□ □□ □□

Algorithm: Q-Learning Routing

1 Q(s, a) □□□ □□□
2 □ h, □□ s(h,k) □□ :
3 a(h,k) <- epsilon-greedy □ □□ □□
4 □□ □□ -> □□ R(h,k) □□
5 □□ □□ s(h+1,k) □□
6 Q □ □□□□ (□□□□ □□ □□)
7 h+1 □ □□□□ □□
8 Q-Table □□ □ □□

Q-Learning □□□ □□ □□ □□

(1) □□ □□

SDN Controller 가
□□□□ □□ ,
□□□ , □□□ □□

(2) □□ □□

Q-Table +
epsilon-greedy □
□□ □□□ □□

(3) □□ □□

Flow Table □□
□□□□ □□
□□□ □□

(4) □□ □□

□□ + □□□ □□
□□ □□ R □□
(□□□ = □□)

(5) Q-Table □□

a, g □ □□□
Q □ □□□□
□□□□ □□

[□□ □□]

□□ Q-Table = □□ □□□ □□
□□□ □□□ Q-Table □ □□ □□□
-> □□ □□ □□□ + QoS □□

□□ □□ □ □□ □□

[□□ □□]

- □□□ 10 □ □□□□ □□□□
- □□ □□□ Ba: 0~1 □□
- □□ : Shortest Path □□□□
- □□ epsilon = 0.3 (5,000 □ □□)

□□ □□
□□□
43.38%
(epsilon=0.3)

□□□ □□
□□□
33.11%
(epsilon=0.3)

□□ □□
□□□
69.57%
(epsilon=0.3)

□□□□	□□ □□	Gain	□□ □□	Gain	□□ □□	Gain
Q-Learning (e=0.3)	**2.01**	**43.38%**	**2.00**	**33.11%**	**4.29**	**69.57%**
Q-Learning (e=0.2)	2.31	34.93%	2.28	23.75%	6.26	55.60%
Q-Learning (e=0.5)	2.08	41.41%	2.07	30.77%	4.40	68.79%
Shortest Path	**3.55**	**-**	**2.99**	**-**	**14.10**	**-**

[□□□] 30% □□ □□□ □□ (delta) □□ □□ □□ -> 까 □□□ □□ □ □□□□□

□□

[□□ □□]

SDN □□□□ □□□ □□□ , □□□ QoS □□□ □□
□□□ □□ □□□ □□□□ Q-Learning □ □□

[□□ □□]

□□ □□□□ □□ □□ □□ 43%, □□ □□ 69% □□
-> □□□ QoS □□ □□

[□□□]

30% □□ □□□ □□□□ □□ □□
-> 가 가 □□ □□ □ □□□□

[□□]

RL □ SDN □□□□ □□ □□□ □□
-> □□□ □□ □□□□ □□□ □□ □□

Q-Learning □□□□ □□ □□ □□

Q(s, a)

Q □ : □□ s □□ □□ a □ □□ □□

$$Q(s, a) \leftarrow Q(s, a) + a\,[\,R + g * \max Q(s', a') - Q(s, a)\,]$$

α (□□)

□□□ : □ □□□ Q 가ㅅ 가 □□ □□□ □□□

R

□□ : □□ □□ □□ □□□ □□□

Q(s, a) □ α (□□□) □□□□

[Q(s, a) - Q □]

□□ :
□□ s □□ □□ a □ □□□ □
□□□ □□ □□ □□

□□ : -∞ ~ +∞

□□ (□□□):
Q(□ 1→ □ 2) = 5.3 (□□ □□)
Q(□ 1→ □ 3) = 2.1 (□□ □□)
Q(□ 1→ □ 4) = -1.0 (□□ □□)

□□ :
" 가 □□□ □□□ □?□□
□□□ □□□ □

[α (□□ , □□□)]

□□ : 0 < α < 1

□□ :
□□□ □□□ □□ Q □□
가 □□ □□□ □?□□

□□ :
α=0.1: Q_new=Q+0.1× □□ (□□)
α=0.3: Q_new=Q+0.3× □□ (□□)
α=0.5: Q_new=Q+0.5× □□ (□□)

□□□□ : □□□□□□ □□
□□□ : □□□□ □□□

R (□□) □ γ (□□□□)

[R - □□ (Reward)]

□□ :
□□ a □ □□ □□
□□□□□□ □□ □□□

□□ : -∞ ~ +∞
□□ -1 ~ +1 □ □□□

□□□ □□ □□ :
R = β1(1-delay) + β2(Bmin/Ba)

□□ :
" □ □□□ □□□□ " □
□□□□□ □□□

[γ (□□ , □□□□)]

□□ : 0 < γ < 1
□□ : 0.9 ~ 0.99

□□ :
□□ □□□ □□ □□□□
가 □ □□□ □□□□

□□ :
γ=0.5: □□□ 50% 가
γ=0.9: □□□ 90% 가

□□□□ : □□ □□ □□ (□□□)
□□□ : □□□ □□ □□ (□□□)

TD Error □ Q □ □□

Temporal Difference (□□□□) Error

TD Error = R + g * max Q(s', a') - Q(s, a)

= [□□ □□ □□ + □□ □□ □□] - [□□ □□□]

TD Error > 0

□□□□ □□

Q □□ □□

TD Error = 0

□□□ □□

Q □ □□ □□

TD Error < 0

□□□□ □□

Q □□ □□

[Q □ □□ (Convergence)]

□□ □□ : TD Error □ □□ -> Q 가ㅅ □□ □□ (α × □ □□)

□□ □□ : TD Error 가 □□ □□□ -> Q □□ □□□ □□

□□ □□ : TD Error ≈ 0 -> Q □□ □□ □□□ □□ (□ □□ □□ □□ □□)

-> □□□□□ Q 가ㅅ □□□□ = □□ □□ = □□ □□□ □□ □□ !